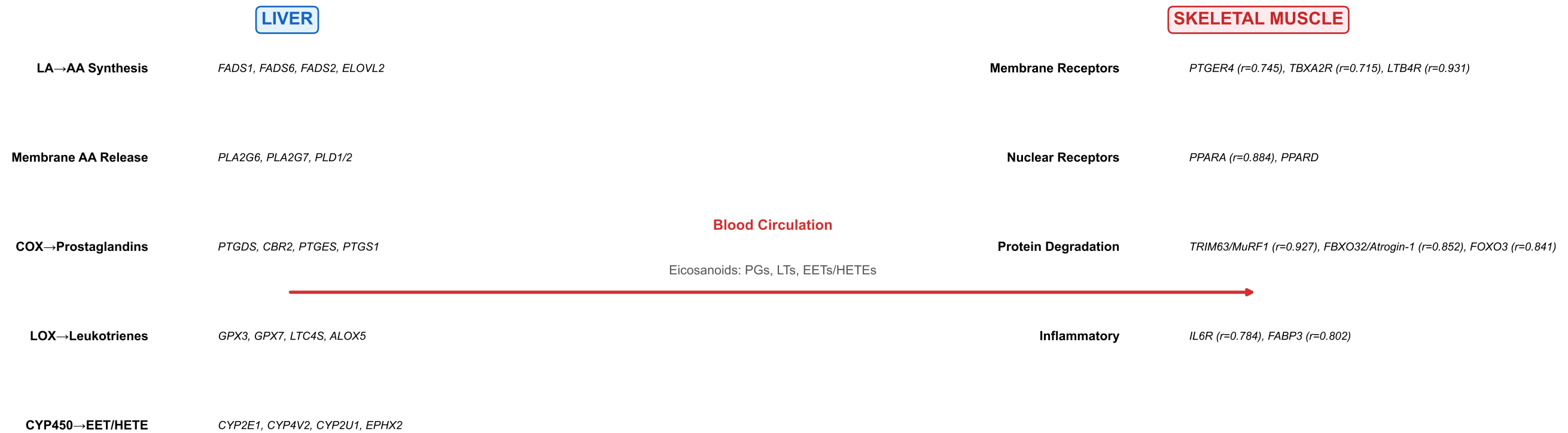


Arachidonic Acid Metabolism — Liver-Muscle Axis Crosstalk Model



Key Finding: Liver AA pathway enzymes show strong positive cross-tissue correlations ($r>0.7$, $P<0.05$) with muscle protein degradation genes (TRIM63, FBXO32) and transcription factors (FOXO3, PPARA). The 'COX→Prostaglandin' and 'CYP450→EET' categories show the strongest coupling (mean $r=0.77-0.82$). This suggests a liver→muscle signaling axis where AA-derived eicosanoids influence muscle proteolysis programs.